

SEMICONDUCTOR MANUFACTURING

Strategy	Description
Investment in semiconductor fabs	Prioritise investments in fabs specialising in automotive-grade semiconductors, meeting high-reliability standards for automotive applications, and focusing on compound semiconductor fabrication and testing facilities
Partnerships with global semiconductor companies	Establish joint ventures or partnerships with global automotive semiconductor companies to gain access to advanced manufacturing technologies, expertise, and supply chains
Focus on specialised automotive semiconductors	Identify and focus on niches within the automotive semiconductor market where India can develop a competitive edge, like EV battery management systems, ADAS chips, or connectivity modules
R&D and innovation ecosystems	Invest in R&D to foster innovation in automotive semiconductor technology, supporting universities, startups, and research institutes. Many of these activities are already in place. Now, the focus should be on semiconductor-driven automotive
Skilled workforce development	Develop specialised education and training programs to create a skilled workforce in semiconductor manufacturing that caters to automotive electronics
Government incentives and policies	Implement favourable policies and incentives to encourage investment in automotive semiconductor manufacturing, including tax breaks and subsidies for R&D. Further enhance the compound semiconductor-based incentives
Supply chain and logistics support	Develop a robust supply chain and logistics infrastructure for automotive semiconductor manufacturing, including raw materials and advanced equipment
Focus on sustainability	Incorporate sustainability into the semiconductor manufacturing process, aligning with global automotive sustainable standards
Market alignment and forecasting	Continuously monitor global and domestic automotive (and semiconductor) markets to align manufacturing capabilities with current and future demand
International collaboration	Engage in international automotive semiconductor collaborations and standard-setting bodies to ensure interoperability of Indian-made automotive semiconductors in the global market

capturing a larger share of domestic and international markets.

Addressing these challenges requires a multi-faceted approach. Indian manufacturers and suppliers must innovate in semiconductor technology development while finding ways to manage costs effectively. It could involve achieving economies of scale, improving operational efficiencies, and forming strategic partnerships to secure better pricing and supply chain reliability. Moreover, the industry is looking towards alternative materials and designs, such as compound semiconductors like silicon carbide (SiC) and gallium nitride (GaN), which offer better performance and could lead to cost savings over the long term despite higher initial investment. To help this cause, the government of India has already laid out an incentive plan to set up compound semiconductor-based semiconductor

fabrication facilities in India.

Furthermore, the strategic importance of semiconductors in India's automotive growth underscores the need for investment in domestic semiconductor manufacturing capabilities. Developing a local semiconductor industry could reduce import dependence, mitigate supply chain risks, and support the automotive sector's ambition to become a global leader.

In summary, the escalating cost of semiconductor content underscores electronics' critical role in the modern automotive industry, particularly for India's growth ambitions. As the country advances towards more connected, autonomous, and electrified vehicles, effectively managing semiconductor costs and bolstering domestic production will be vital to sustaining innovation and competitiveness in the global automotive landscape.

How India can speed up automotive growth with semiconductor manufacturing

India can strategically plan semiconductor manufacturing to cater to its automotive industry demand through several logical steps as shown in the table.

By systematically addressing these areas, India can effectively plan and establish a semiconductor manufacturing ecosystem that supports its every-green automotive industry, ensuring self-reliance and a strong position in the global automotive market.

Future course of automotive semiconductor in India

The future course of automotive semiconductors in India is poised at an exciting juncture. Converging several key trends and strategic initiatives drives significant growth and innovation potential.

As the Indian automotive industry continues to evolve towards electrification, connectivity, and autonomy, the role of semiconductors will become increasingly central, offering a unique opportunity for India to establish itself as a critical player in the global automotive semiconductor market.

In conclusion, the future course of automotive in India is marked by opportunities for growth, innovation, and global leadership in semiconductor. By focusing on semiconductor design and manufacturing that caters to new-age automotive requirements, India can navigate the complexities of the worldwide semiconductor market and carve out niche self-reliance in the automotive sector, too.

The journey will require coordinated efforts across the government, industry, and academia, but the potential rewards for economic growth, technological advancement, and strategic autonomy are substantial. **EFY**